数学笔记

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2025 年 12 月 5 日

目录

Reference: Theory of Orlicz spaces, by Rao and Ren. Application of Orlicz spaces, by Rao.

A function Φ is called a **Young function** if $\Phi(0) = 0$, Φ is even, convex, strictly increasing on $[0, \infty)$ and $\lim_{t\to\infty} t^{-1}\Phi(t) = \infty$. The convexity of Φ and $\Phi(0) = 0$ implies that $\Phi(cx) \leq c\Phi(x)$ and $\Phi(c^{-1}x) \geq c^{-1}\Phi(x)$ for all $c \in (0, 1)$, $x \in \mathbb{R}$. The dual function of Φ is Ψ defined by $\Psi(u) := \sup_v\{u \cdot v - \Phi(v)\}$.

$$\widetilde{L}^{\Phi} = \left\{ f : \Omega \to \mathbb{R}, \text{measurable}, \rho_{\Phi}(f) = \int_{\Omega} \Phi(f)\,\mathrm{d}\mu < \infty \right\}$$

The space

$$L^{\Phi} = \{f : \Omega \to \mathbb{R}, \text{measurable}, \rho_{\Phi}(af) < \infty, \text{for some } a > 0\}$$

is called an Orlicz space on (Ω, Σ, μ).

The space

$$M^{\Phi} = \{f : \Omega \to \mathbb{R}, \text{measurable}, \rho_{\Phi}(af) < \infty, \text{for all } a > 0\}$$

is called the Morse-Transue space.

The **Orlicz norm** is

$$\|f\|_{\Phi} := \sup \left\{ \int_{\Omega} |fg|\,\mathrm{d}\mu : \rho_{\Psi}(g) \leq 1 \right\}$$

The **gauge norm** is

$$\|f\|_{(\Phi)} := \inf\left\{k \geq 0 : \rho_\Phi\left(\frac{f}{k}\right) \leq 1\right\}$$

where (Φ, Ψ) are complementary pair of Young functions and $(L^\Phi(\mu), L^\Psi(\mu))$ are the corresponding Orlicz spaces on a measure space (Ω, Σ, μ).

Also, $(L^\Phi, \|\cdot\|_\Phi)$ is a Banach space if f, g are identified when $\|f-g\|_\Phi = 0$.

Under some good condition, the dual space of $L^\Phi(\mu)$ is $L^{(\Psi)}(\mu)$. We have Holder inequality:

$$\|fg\|_1 \leq \|f\|_{(\Phi)}\|g\|_\Psi \qquad \|fg\|_1 \leq \|f\|_\Phi\|g\|_{(\Psi)}$$

for $f \in L^\Phi, g \in L^\Psi$.

Define

$$M^\Phi := \{f : \rho_\Phi(\alpha f) < \infty, \forall \alpha > 0\}$$

If Φ is Young function, then $M^\Phi \subset \widetilde{L}^\Phi \subset L^\Phi$.

For every $0 \neq f \in M^\Phi$, we have

$$\int_\Omega \Phi\left(\frac{f}{\|f\|_{(\Phi)}}\right) \mathrm{d}\mu = 1$$

注 0.1. For simplicity, we use L^Φ and $L^{(\Phi)}$ for spaces $(L^\Phi, \|\cdot\|_\Phi)$ and $(L^\Phi, \|\cdot\|_{(\Phi)})$ respectively. Similar for M^Φ and $M^{(\Phi)}$. In case μ is counting measure, the corresponding spaces are denoted $\ell^\Phi, \ell^{(\Phi)}$ and $m^\Phi, m^{(\Phi)}$ respectively.

Theorem 12. *Let Φ, Ψ be a pair of complementary N-functions. Then for any measurable function f on (Ω, Σ, μ), we have*

$$\|f\|_{(\Phi)} = \sup\left\{\left|\int_\Omega fg\ d\mu\right| :\ g \in M^\Psi, \|g\|_\Psi \leq 1\right\}, \tag{16}$$

and

$$\|f\|_\Phi = \sup\left\{\left|\int_\Omega fg\ d\mu\right| :\ g \in M^{(\Psi)}, \|g\|_{(\Psi)} \leq 1\right\}. \tag{17}$$

The dual of L^Φ with the Luxemburg norm is L^Ψ with the Orlicz norm (under certain conditions, e.g., Φ and Ψ satisfy the Δ_2 -condition for finite measures).

The question is whether the following are dual norms.

For $p, q \in (\Delta_n, \oplus; \mathbb{R}, \odot)$,

$$\|p\|_{(\Phi)} := \inf\left\{k > 0 : \sum_{i=0}^{n} \Phi\left(\frac{\mathrm{clr}_i(p)}{k}\right) \leq 1\right\}$$

where $\mathrm{clr}(p) = \left[\log \frac{p_0}{\left(\prod_{i=0}^{n} p_i\right)^{1/(n+1)}}, \ldots, \log \frac{p_n}{\left(\prod_{i=0}^{n} p_i\right)^{1/(n+1)}}\right]$.

$$\ell_p(q) := \sum_{i=0}^{n} \mathrm{clr}_i(p)\mathrm{clr}_i(q)$$

We define a norm on Δ_n^*, the linear functional space,

$$\|\ell_p\|_\Psi := \sup\left\{\langle p, q\rangle : \rho_\Phi(q) = \sum_{i=0}^{n} \Phi(\mathrm{clr}_i(q)) \leq 1\right\} \overset{\text{Theorem 8}}{=} \underbrace{\sup\{\langle p, q\rangle : \|q\|_{(\Phi)} \leq 1\}}_{=\|p\|_{\Phi,*}}$$

where $\langle p, q\rangle := \ell_p(q)$.

定理 0.2. *$(\Delta_n, \|\cdot\|_{(\Phi)})$ and $(\Delta_n^*, \|\cdot\|_\Psi)$ are dual space.*

证明. Omitted. □

Theorem 8. *Let Φ, Ψ be a complementary pair of N-functions and L^Φ, L^Ψ be the corresponding Orlicz spaces on (Ω, Σ, μ). Then*

(i) for $0 \neq f \in L^\Phi$, $\rho_\Phi(f) \leq 1$ iff $\|f\|_{(\Phi)} \leq 1$ so that

$$\|f\|_{(\Phi)} > 1 \Rightarrow \rho_\Phi(f) \geq \|f\|_{(\Phi)} > 1;$$

(ii) $\|f\|_{(\Phi)} \leq \|f\|_\Phi \leq 2\|f\|_{(\Phi)}, f \in L^\Phi$, (10)

so that these are equivalent (norm) functionals;

(iii) (Improved Hölder's inequality) $f \in L^\Phi$, $g \in L^\Psi \Rightarrow fg \in L^1$ and

$$\|fg\|_1 \leq \|f\|_{(\Phi)}\|g\|_\Psi;\ (\|fg\|_1 \leq \|f\|_\Phi\|g\|_{(\Psi)}). \tag{11}$$

In the case $\Phi(u) = |u|^p, 1 < p < \infty$, we have $\|\cdot\|_{(\Phi)} = \|\cdot\|_p$ and $\|\cdot\|_\Psi = \|\cdot\|_q$.

Example 10. If $\Phi_p(u) = |u|^p, 1 < p < \infty$, then the complementary N-function Ψ is seen to be with $q = \frac{p}{p-1}$,

$$\Psi_q(v) = \left(\frac{q-1}{q}\right)^{q-1} \frac{|v|^q}{q}.$$

Then $k_0 = \|f\|_p < \infty$, is the Lebesgue L^p-norm, and

$$\|f\|_{(\Phi_p)} = \inf\left\{k > 0 : \rho_{\Phi_p}\left(\frac{f}{k}\right) \leq 1\right\}.$$

The infimum is attained for $k = k_0$ so that $\|f\|_{(\Phi_p)} = \|f\|_p$. However, the same is not true for $\|\cdot\|_{(\Psi_q)}$, but we have for $g \in L^{\Psi_q}$,

$$\begin{aligned} \|g\|_{\Psi_q} &= \sup\left\{\int_\Omega |fg| d\mu : \|f\|_{(\Phi_p)} \leq 1\right\} \\ &= \sup\left\{\int_\Omega |fg| d\mu : \|f\|_p \leq 1\right\} = \|g\|_q. \end{aligned} \tag{13}$$

The reason for this difference is that (Φ_p, Ψ_q) is not a normalized pair. Thus alternating the gauge and Orlicz norms we get a classical form of some results such as the Hölder inequality.

When $\Phi(u) = u^2/2, \Psi(u) = u^2/2$, this is the inner product case. The norm we defined is a generalization.

Compositional Data Analysis CDA (Aitchison)

$(\Delta_n, \oplus; \mathbb{R}, \odot)$ is referred to Aitchison geometry in the "compositional data analysis" community.

- Centered log-ratio (clr) of p is

$$\mathrm{clr}(\mathrm{p}) = \left[\log \frac{p_0}{(\Pi_{i=0}^n p_i)^{1/(n+1)}}, \cdots, \log \frac{p_n}{(\Pi_{i=0}^n p_i)^{1/(n+1)}}\right]$$

  so it satisfies $\mathrm{clr}(\kappa \mathrm{p}) = \mathrm{clr}(\mathrm{p})$. The inverse is the soft-max representation.

- Inner product $\langle \cdot, \cdot \rangle$ can be introduced (prove!):

$$\langle \mathrm{p}, \mathrm{q} \rangle = \sum_{i=0}^n \mathrm{clr}_i(\mathrm{p})\mathrm{clr}_i(\mathrm{q}) = \frac{1}{2(n+1)} \sum_{i=0}^n \sum_{j=0}^n \log \frac{p_i}{p_j} \log \frac{q_i}{q_j}.$$